Limiting cases ($\Lambda \to 0$, $\vee \to \infty$) of the finite terms: cancellation of the negative-argument logarithms, reality, and simplification

1 Where an i can come from

In the limiting-case file the possible sources of an imaginary part are:

(C1) an explicit i in the coefficient ($ig^4 f^{abd}$, $ig^4 f^{c'd'a}$, ...);

(C2) the Fourier phases $e^{-ik\cdot(w'-w)}$, $e^{-ik\cdot(y'-w)}$, $e^{-ik\cdot(z-x)p^+/k^+}$, ...;

(C3) the momentum integral of Group I, Row II (N_{2b});

(C4) logarithms whose argument can be negative. In the limiting case these occur *only* in Group I, Row III.

All Wilson lines and f^{abc} are real, ρ is hermitian, every transverse kernel is a real rational function, and $\mathcal{A}_0$ is real. Its radicand $(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2 \geq 0$ by Cauchy–Schwarz.

Rule. Groups I and III are interferences $\mathcal{A}_{\rm LO}^\dagger \mathcal{A}_{\rm NLO}$ and come with their c.c.; Group II is $|\mathcal{A}|^2$ and must be hermitian by itself. For a term $T = \int e^{-ik\cdot\Delta}\, c\, \mathcal{K}\, \mathcal{O}$ with real kernel $\mathcal{K}$ and classical ρ:

$$c \text{ real:} \quad T + T^\dagger = 2c \int \cos(k\cdot\Delta)\, \mathcal{K}\, \mathcal{O}, \qquad c = i\alpha: \quad T + T^\dagger = 2\alpha \int \sin(k\cdot\Delta)\, \mathcal{K}\, \mathcal{O}, \tag{1.1}$$

because $i\, e^{-ik\cdot\Delta} + (i\, e^{-ik\cdot\Delta})^* = 2\sin(k\cdot\Delta)$.

2 Logarithms in the limiting-case file

Row	logarithms	argument
GI R2 (N_{2b})	$\log \frac{p^2}{k^2}$, $\log \frac{k^2}{(k-p)^2}$	positive
GI R3	$\log \frac{(x-z)^2}{(x-z)^2-(x-w)^2}$, $\log \frac{(x-w)^2-(x-z)^2}{(x-w)^2}$, $\log \frac{-(x-z)^2}{(x-w)^2}$, $\log \frac{-(x-z)^2}{(x-w)^2-(x-z)^2}$	can be negative
GI R3 (N_{3b})	$\log \frac{(z-w)^2}{(x-w)^2}$	positive
GI R4, GII R1, GIII R1, R2	$\log \frac{(x-z)^2}{(x-w)^2}$, $\log \frac{(x'-z)^2}{(x'-w')^2}$, ...	positive
GII R1 ($N_{2d,2e,2f}$)	$\log \frac{(x-z)^2}{(x-w)^2} / [(x-z)^2 - (x-w)^2]$	positive; the pole is removable (the log vanishes there)

So the only logarithms with negative arguments are in Group I, Row III. Sec. 3 treats them in full.

3 Group I, Row III

3.1 Notation

$$X \equiv x-w, \quad Z \equiv x-z, \quad W \equiv z-w = X-Z, \quad a \equiv x'-w', \quad n \equiv y-z, \quad u \equiv \frac{Z^2}{X^2} = \frac{(x-z)^2}{(x-w)^2} > 0. \tag{3.1}$$

All four pieces N_1, N_2, N_{3a}, N_{3b} of Row III have the *same* colour structure

$$\mathcal{C}_{\rm III} = \Big[U^{ab'}(x') - U^{ab'}(w')\Big] \rho^{b'}(x') \Big[U^{de}(x)\, U^{bc}(z) - U^{de}(x)\, U^{bc}(y)\Big] \rho^e(x)\, \rho^c(y), \tag{3.2}$$

and the same overall factor. So the whole row can be written as one kernel:

$$\frac{d^3 N_{\rm III}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{ig^4 f^{abd}}{16\pi^5} \frac{1}{k^+} \int_{w,w'} e^{-ik\cdot(w'-w)} \int_{x,y,z,x'} \frac{a^i\, n^j}{a^2\, n^2}\, M^{ij}(x,z,w)\, \mathcal{C}_{\rm III}. \tag{3.3}$$

Terms written with $\frac{ig^4}{8\pi^5}$ contribute with a factor 2 to M^{ij}.

3.2 Step 1: the branch of the logarithms

The logs come from p^+ integrals with the energy denominator $p^+(x-z)^2 - k^+(x-w)^2$ of the single-gluon state $\bar{\mathbb{B}}_3$. In $\bar{\mathbb{B}}_3^\dagger$ this denominator carries $-i0$. That is exactly the *principal branch*, $\log(-A) = \log A + i\pi$ for $A > 0$, which was checked earlier at finite $\vee$ against the residue sum to 12 digits. The limit $\vee \to \infty$ ($\kappa = k^+/\vee \to 0$) does not change the branch. So no absolute values are needed.

3.3 Step 2: every log on the principal branch

With $\theta(\cdot)$ the step function:

$$L_{1a} \equiv \log\frac{Z^2}{Z^2 - X^2} = \log\frac{u}{u-1} = \log\frac{u}{|1-u|} + i\pi\,\theta(1-u), \tag{3.4}$$

$$L_{1b} \equiv \log\frac{X^2 - Z^2}{X^2} = \log(1-u) = \log|1-u| + i\pi\,\theta(u-1), \tag{3.5}$$

$$L_{1c} \equiv \log\frac{-Z^2}{X^2} = \log(-u) = \log u + i\pi, \tag{3.6}$$

$$L_2 \equiv \log\frac{-Z^2}{X^2 - Z^2} = \log\frac{-u}{1-u} = \log\frac{u}{|1-u|} + i\pi\,\theta(1-u), \tag{3.7}$$

$$L_3 \equiv \log\frac{X^2 - Z^2}{X^2} = \log(1-u) = \log|1-u| + i\pi\,\theta(u-1). \tag{3.8}$$

Adding the pairs that appear together, the $\log|1-u|$ and the step functions cancel:

$$\boxed{L_{1a} + L_{1b} = L_2 + L_3 = L_{1c} = \log u + i\pi = \log(-u) \qquad \text{for all } u > 0.} \tag{3.9}$$

For example, for $u < 1$: $L_2 + L_3 = \big[\log\frac{u}{1-u} + i\pi\big] + \log(1-u) = \log u + i\pi$. For $u > 1$: $L_2 + L_3 = \log\frac{u}{u-1} + \big[\log(u-1) + i\pi\big] = \log u + i\pi$.

3.4 Step 3: collecting the logarithmic terms

N_1: the first two terms have the same kernel, so they give $(L_{1a} + L_{1b})$. The third term has L_{1c}:

$$M^{ij}_{N_1} = \frac{X^2 + Z^2}{(Z^2 - X^2)^2}\,\delta^{ij}\,(L_{1a} + L_{1b}) - \frac{2X^iZ^j}{X^2Z^2}\,L_{1c}. \tag{3.10}$$

N_2 (all with L_2, overall sign $-$, factor 2) and the first six terms of N_{3a} (all with L_3, factor 2) contain the *same six tensors*:

tensor	N_2	N_{3a}
$\delta^{ij}\frac{(W\cdot X)Z^2}{W^2(X^2-Z^2)^2}$	$-2L_2$	$-2L_3$
$\delta^{ij}\frac{(W\cdot Z)X^2}{W^2(X^2-Z^2)^2}$	$+2L_2$	$+2L_3$
$\frac{W^iX^j}{W^2(X^2-Z^2)}$	$-2L_2$	$-2L_3$
$\frac{X^iW^jZ^2}{W^2(X^2-Z^2)X^2}$	$+2L_2$	$+2L_3$
$\frac{W^iZ^jX^2}{W^2(X^2-Z^2)Z^2}$	$+2L_2$	$+2L_3$
$\frac{Z^iW^j}{W^2(X^2-Z^2)}$	$-2L_2$	$-2L_3$

Hence they combine into $(L_2 + L_3)$. Using (3.9), all logarithmic terms of $N_1 + N_2 + N_{3a}$ share the single factor $\log(-u)$:

$$M^{ij}_{\log} = \log(-u)\Bigg\{\frac{X^2+Z^2}{(X^2-Z^2)^2}\,\delta^{ij} - \frac{2X^iZ^j}{X^2Z^2} + \frac{2}{W^2(X^2-Z^2)}\Big[\frac{(W\cdot Z)X^2 - (W\cdot X)Z^2}{X^2-Z^2}\,\delta^{ij}$$
$$- W^iX^j + \frac{Z^2}{X^2}X^iW^j + \frac{X^2}{Z^2}W^iZ^j - Z^iW^j\Big]\Bigg\}. \tag{3.11}$$

3.5 Step 4: the bracket collapses to the Row IV kernel

δ^{ij} part. With $W = X - Z$: $W\cdot Z = X\cdot Z - Z^2$, $W\cdot X = X^2 - X\cdot Z$, $W^2 = X^2 + Z^2 - 2X\cdot Z$, so

$$(W\cdot Z)X^2 - (W\cdot X)Z^2 = (X\cdot Z)(X^2+Z^2) - 2X^2Z^2,$$

$$\frac{(X^2+Z^2)W^2 + 2(X\cdot Z)(X^2+Z^2) - 4X^2Z^2}{W^2(X^2-Z^2)^2} = \frac{(X^2+Z^2)^2 - 4X^2Z^2}{W^2(X^2-Z^2)^2} = \frac{1}{W^2}.$$

Non-δ part. One needs $\frac{1}{X^2-Z^2}\big[-W^iX^j+\frac{Z^2}{X^2}X^iW^j+\frac{X^2}{Z^2}W^iZ^j-Z^iW^j\big]=\frac{W^iZ^j}{Z^2}-\frac{X^iW^j}{X^2}$. Multiply by X^2-Z^2; the difference of the two sides is $-W^i(X-Z)^j+(X-Z)^iW^j=-W^iW^j+W^iW^j=0$. Therefore

$$\boxed{M^{ij}_{\log}=2\,\log(-u)\,\mathcal{K}^{ij}_{\rm IV},\qquad \mathcal{K}^{ij}_{\rm IV}=\frac{\delta^{ij}}{2(z-w)^2}+\frac{(z-w)^i(x-z)^j}{(z-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}-\frac{(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2},}\tag{3.12}$$

which is exactly the kernel of Group I, Row IV. All the $\frac{1}{(x-z)^2-(x-w)^2}$ poles have disappeared. **Numerical check:** for random x,z,w (both $u<1$ and $u>1$), $\max|M^{ij}_{\log}-2(\log u+i\pi)\mathcal{K}^{ij}_{\rm IV}|\le 7\times10^{-13}$, computed from the file's terms on the principal branch.

3.6 Step 5: the $\mathcal{A}_0$ terms

Since $Z^2-X\cdot Z=-Z\cdot W$ and $|X\times Z|=|W\times Z|$, the two arctangents add up to the angle of the triangle (x,z,w) at w:

$$\boxed{\mathcal{A}_0=\frac{\theta_w}{\big|(x-w)\times(z-w)\big|},\qquad \theta_w=\arccos\frac{(x-w)\cdot(z-w)}{|x-w|\,|z-w|}\in[0,\pi],}\tag{3.13}$$

which was checked numerically to 10^{-12}. The four $\mathcal{A}_0$ terms (N_{3a}: 3, N_{3b}: 1) combine, using the projections $X_\perp=X-\frac{(X\cdot Z)}{Z^2}Z$ and $Z_\perp=Z-\frac{(X\cdot Z)}{X^2}X$, into

$$a^in^jM^{ij}_{\mathcal{A}}=\frac{2\,\mathcal{A}_0}{(z-w)^2}\Big[(a\cdot W)\,(n\cdot X_\perp)+(a\cdot Z_\perp)\,(n\cdot W)\Big].\tag{3.14}$$

3.7 Step 6: the remaining $\log\frac{(z-w)^2}{(x-w)^2}$ terms (N_{3b})

$$a^in^jM^{ij}_{B}=\frac{1}{(z-w)^2}\log\frac{(z-w)^2}{(x-w)^2}\Big[-\frac{(a\cdot W)(n\cdot Z)}{Z^2}+\frac{(a\cdot X)(n\cdot W)}{X^2}-a\cdot n\Big].\tag{3.15}$$

3.8 Step 7: Row III, simplified

$$\begin{aligned}\frac{d^3N_{\rm III}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{ig^4f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{w,w'}e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\frac{1}{(x'-w')^2(y-z)^2}\Bigg\{\\
&2\log\frac{(x-z)^2}{(x-w)^2}\,(x'-w')^i(y-z)^j\,\mathcal{K}^{ij}_{\rm IV}\\
&+\frac{2\,\theta_w}{|(x-w)\times(z-w)|\,(z-w)^2}\Big[(x'-w')\cdot(z-w)\;(y-z)\cdot X_\perp+(x'-w')\cdot Z_\perp\;(y-z)\cdot(z-w)\Big]\\
&+\frac{1}{(z-w)^2}\log\frac{(z-w)^2}{(x-w)^2}\\
&\quad\times\Big[-\frac{(x'-w')\cdot(z-w)\;(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(x'-w')\cdot(x-w)\;(y-z)\cdot(z-w)}{(x-w)^2}-(x'-w')\cdot(y-z)\Big]\Bigg\}\mathcal{C}_{\rm III}\\
&+\frac{d^3N_{i\pi}}{d^3k},\end{aligned}\tag{3.16}$$

with the imaginary part of $\log(-u)$, multiplied by the overall i, giving the *real* term

$$\frac{d^3N_{i\pi}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4f^{abd}}{8\pi^4}\,\frac{1}{k^+}\int_{w,w'}e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\,\mathcal{K}^{ij}_{\rm IV}\,\mathcal{C}_{\rm III}.\tag{3.17}$$

(At finite $\vee$ this carried $\theta(u-\kappa)$; for $\vee\to\infty$, $\theta(u)=1$.)

3.9 Step 8: adding the c.c.

The first three lines of (3.16) have coefficient $i\times$real, so they are class $i\alpha$ with $\alpha=\frac{g^4f^{abd}}{16\pi^5k^+}$. $N_{i\pi}$ has a real coefficient. By (1.1), for classical ρ:

$$\begin{aligned}\frac{d^3N_{\rm III}}{d^3k}+\text{c.c.}=&\frac{1}{(2\pi)^3}\,\frac{g^4f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int\sin\big(k\cdot(w'-w)\big)\Big\{\text{first three lines of (3.16)}\Big\}\mathcal{C}_{\rm III}\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4f^{abd}}{4\pi^4}\,\frac{1}{k^+}\int\cos\big(k\cdot(w'-w)\big)\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\,\mathcal{K}^{ij}_{\rm IV}\,\mathcal{C}_{\rm III},\end{aligned}\tag{3.18}$$

which is manifestly real. No log of a negative number and no explicit i is left.

3.10 Step 9: singularities

- $(x-z)^2 = (x-w)^2$**.** All such poles cancelled in Step 4.
- $z \to y$**,** $x \to z$**,** $x \to w$**.** All are integrable: at most $1/|\cdot|$ or $\log|\cdot|/|\cdot|$.
- $z \to w$**, in** $N_{i\pi}$**.** $\mathcal{K}_{\rm IV} \to \frac{\delta^{ij}}{2(z-w)^2}$, which is log divergent. At $z=w$ the colour structure becomes $f^{abd}X^a(w')\cdot Y^b(w)\,D^d$, with $X^a(w') = \int_{x'} \frac{(x'-w')}{(x'-w')^2}[U(x')-U(w')]^{ab'}\rho^{b'}(x')$, $Y^b(w) = \int_y \frac{(y-w)}{(y-w)^2}[U(w)-U(y)]^{bc}\rho^c(y)$ and $D^d = \int_x U^{de}(x)\rho^e(x)$. This is antisymmetric under $w \leftrightarrow w'$, and $N_{i\pi}$ has a real coefficient, so by (1.1) its divergent part is multiplied by $\cos(k\cdot(w'-w))$, which is symmetric, and vanishes. It is a **Coulomb phase**, and it cancels exactly against the c.c. The rest of $N_{i\pi}$ is finite.
- $z \to w$**, in the** $\log u$ **and** $\mathcal{A}_0$ **terms.** Both are finite: $\log u = O(|z-w|)$ is odd in $z-w$, and $\mathcal{A}_0$ times its bracket is $O(1/|z-w|)$. Numerically, $r^2\times$(angle average) goes to 0 as $r=|z-w|\to 0$.
- $z \to w$**, in the last term of** N_{3b}: $-\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\frac{1}{(z-w)^2}\log\frac{(z-w)^2}{(x-w)^2}$. This behaves like $\frac{\log r^2}{r^2}$, a *double-log* divergence of $\int d^2z$. Numerically, $r^2\times$(angle average) $= 2.98, 5.10, 7.22, 9.34$ for $r = 10^{-1},\dots,10^{-4}$: it grows by $\approx 2\log 10$ per decade. It is class $i\alpha$, so with the c.c. it multiplies $\sin(k\cdot(w'-w))$, which is antisymmetric, times the antisymmetric colour structure above. The product is symmetric, so it does *not* cancel against the c.c. Nothing else in Row III has this $z\to w$ behaviour.

Consequence. With this one exception, Row III in the limit $\Lambda\to 0$, $\vee\to\infty$ is finite and real. The $\frac{1}{(z-w)^2}\log\frac{(z-w)^2}{(x-w)^2}$ term of N_{3b} cannot be taken to $\Lambda\to 0$ on its own. It is the remainder of a p^+ integral whose $\log(k^+/\Lambda)\frac{1}{(z-w)^2}$ partner went into the JIMWLK file, and the two must be combined, with the same colour structure, before $\Lambda\to 0$. Equivalently, the $z\to w$ region must be cut off at the scale set by Λ, e.g. $(z-w)^2 \gtrsim \frac{\Lambda}{k^+}(x-w)^2$. The origin of this term needs to be rechecked.

4 The other rows

Row	class	how it is real
GI R2, N_{2b}	$i\alpha$ (implicit)	$\Phi^i = \int d^2q\,d^2p\,e^{iq\cdot(w-y)+ip\cdot(y-x)}\frac{R^i(q,p)}{q^2p^2}$ is $i\times$real (R^i is odd under $(q,p)\to(-q,-p)$). With c.c.: sin. Rename the dummy $\int d^2k \to \int d^2q$: it clashes with the external k.
GI R4	$i\alpha$	positive log; with c.c.: $\frac{g^4}{4\pi^5}\sin(k\cdot(w'-w))$
GI R5, first part	$i\alpha$	phase $e^{-ik\cdot(w'-Y_p)}$ with real $Y_p = (1-\frac{p^+}{k^+})z + \frac{p^+}{k^+}x$; with c.c.: sin
GI R5, N_{2a}, $N_{2b\,\rm rest}$	real	with c.c.: cos
GII R1, first part	—	$N_{1b} = N_{1a}^\dagger$, so $N_{1a}+N_{1b} = -\frac{g^4}{8\pi^5 k^+}\int\sin(k\cdot(w'-w))(\dots)$
GII R1, N_{2a}–N_{2f}	real	each kernel is even under $(x,w)\leftrightarrow(x',w')$; hermitian, cos
GII R4	real	symmetric under $(x,y)\leftrightarrow(x',y')$; $\cos(k\cdot(y'-y)/\xi)$
GIII R1, R2, R3 (first), R5, R6	$i\alpha$	positive logs and denominators; with c.c.: sin
GIII R3 (second)	real	with c.c.: cos

5 Summary

1. **Negative-argument logs (only GI R3).** On the principal branch, which is the physical $-i0$ of $\bar{\mathbb{B}}_3^\dagger$, they combine *exactly* into one factor $\log(-u) = \log\frac{(x-z)^2}{(x-w)^2} + i\pi$, multiplying $2\mathcal{K}_{\rm IV}$. All the $\log|1-u|$ pieces and step functions cancel, and so do all the $\frac{1}{(x-z)^2-(x-w)^2}$ poles.
2. **The** $i\pi$**.** Times the overall i it becomes the *real* term $N_{i\pi}$. Its only divergence ($z\to w$) is a Coulomb phase, which cancels against the c.c.
3. **Explicit** i**'s and phases.** They give sin (class $i\alpha$) or cos (real class) with the c.c. in Groups I and III. Group II is hermitian by itself. The final cross section is real.
4. **Simplifications.** Row III is eq. (3.16), with $\mathcal{A}_0 = \theta_w/|(x-w)\times(z-w)|$.
5. **Open issue.** The last term of N_{3b} has a non-integrable $\frac{\log(z-w)^2}{(z-w)^2}$ at $z\to w$ that survives the c.c. It must be recombined with its $\log(k^+/\Lambda)$ partner before $\Lambda\to 0$.